

Steps

Reason

1.  $S(\text{Doug}) \wedge J(\text{Doug})$  Premise
2.  $\forall x (J(x) \Rightarrow \text{Job}(x))$  Premise
3.  $S(\text{Doug})$  Simplification on 1
4.  $J(\text{Doug}) \wedge S(\text{Doug})$  Commutative Law on 1
5.  $J(\text{Doug})$  Simplification on 4
6.  $J(\text{Doug}) \Rightarrow \text{Job}(\text{Doug})$  Universal instantiation on 2
7.  $\text{Job}(\text{Doug})$  Modus ponens on 5, 6
8.  $S(\text{Doug}) \wedge \text{Job}(\text{Doug})$  Conjunction on 3, 7
9.  $\exists x (S(x) \wedge \text{Job}(x))$  Existential generalization on 8

b) "Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution."

Soln:  $C(x)$ :  $x$  is a person of this class  
 $O(x)$ :  $x$  cares about ocean pollution.  
 $W(x)$ :  $x$  enjoys whale watching  
Domain: Set of all people in the world.

Premise: 1)  $\exists x (C(x) \wedge W(x))$  2)  $\forall x (W(x) \Rightarrow O(x))$

Conclusion:  $\exists x (C(x) \wedge O(x))$

Steps

Reason

1.  $\exists x (C(x) \wedge W(x))$  Premise
2.  $\forall x (W(x) \Rightarrow O(x))$  Premise
3.  $C(p) \wedge W(p)$  for some  $p$  Existential instantiation on 1
4.  $W(p) \wedge C(p)$  Commutative Law on 3
5.  $C(p)$  Simplification on 3
6.  $W(p)$  Simplification on 4